

KOCHI, India

WMO: 433530

Lat: 9.946N

Lon: 76.272E

Elev: 8

StdP: 14.69

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	73.0	73.9	62.0	83.0	83.0	63.9	89.0	82.0	21.6	78.0	17.7	79.8	3.1	320	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.2	91.7	80.2	90.9	79.8	90.1	79.4	82.4	89.1	81.7	88.4	81.1	87.8	10.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	158.5	86.7	79.8	154.7	86.1	79.2	151.4	85.6	46.2	89.3	45.4	88.7	44.8	88.3	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	14.4	13.0	DB	68.8	95.8	2.3	3.6	67.2	98.4	65.8	100.4	64.5	102.4	62.9	105.0	(4)
(5)			WB	65.0	84.6	2.0	1.0	63.6	85.4	62.4	86.0	61.2	86.6	59.8	87.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	81.6	82.8	84.4	84.8	84.1	81.0	80.1	80.2	81.0	81.4	81.8	81.7	
	DBStd	2.34	1.41	1.55	1.55	1.83	2.30	2.39	1.94	1.93	1.59	1.61	1.39	1.59	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	11704	979	918	1066	1043	1056	931	934	937	929	972	955	984	
	CDD65	6229	514	498	601	593	591	481	469	472	479	507	505	519	
	CDH74	68606	5497	5762	7649	7704	7499	4813	4232	4345	4803	5232	5448	5622	
	CDH80	24273	1926	2181	3358	3601	3311	1410	975	965	1244	1563	1795	1944	
Wind	WSAvg	5.1	4.6	4.8	5.2	5.6	5.6	5.2	5.3	5.6	5.6	4.9	4.1	4.3	
Precipitation	PrecAvg	127.6	1.2	0.8	1.5	4.5	13.5	26.5	29.9	16.3	12.6	11.4	7.2	2.1	
	PrecMax	186.5	6.7	6.7	5.5	11.5	33.5	44.6	57.2	25.4	23.1	20.2	18.7	10.9	
	PrecMin	89.1	0.0	0.0	0.0	1.4	1.0	8.6	18.0	7.2	2.9	3.5	0.7	0.0	
	PrecStd	20.4	1.9	1.5	1.4	2.2	11.0	10.2	9.7	5.5	6.1	4.6	4.8	3.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.3	91.5	92.4	93.1	92.7	90.0	88.7	88.0	88.6	89.3	90.0	90.4	
		MCWB	74.6	76.1	79.8	81.2	81.4	80.6	80.3	79.5	79.4	79.0	78.0	76.5	
	2%	DB	89.1	89.9	91.0	91.7	91.3	88.8	87.2	86.5	87.2	88.1	88.9	89.1	
		MCWB	74.9	76.1	79.3	80.8	81.0	80.2	79.5	78.9	78.6	78.8	77.9	76.3	
	5%	DB	88.1	89.0	90.1	90.9	90.4	87.6	86.0	85.6	86.3	87.2	88.0	88.1	
		MCWB	74.7	76.1	79.1	80.6	80.8	79.8	79.1	78.5	78.4	78.6	77.9	76.0	
	10%	DB	87.2	88.0	89.3	90.0	89.5	86.2	84.8	84.6	85.5	86.2	87.0	87.2	
		MCWB	74.7	76.1	78.9	80.4	80.5	79.4	78.6	78.2	78.1	78.3	77.8	75.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.6	80.3	82.4	83.4	83.4	82.2	81.5	80.9	80.9	81.0	81.0	80.4
			MCDB	85.8	87.8	89.1	90.0	89.8	87.7	86.7	86.0	86.3	86.8	87.4	87.5
2%		WB	78.3	79.2	81.0	82.2	82.3	81.3	80.5	79.8	79.9	80.0	80.0	79.1	
		MCDB	85.1	86.4	88.6	89.5	89.1	86.7	85.7	84.9	85.4	85.9	86.5	86.0	
5%		WB	77.5	78.5	80.4	81.6	81.6	80.6	79.6	79.1	79.1	79.2	79.2	78.3	
		MCDB	84.3	85.5	88.1	88.9	88.4	85.9	84.5	84.0	84.5	85.1	85.5	85.0	
10%		WB	76.9	78.0	79.8	80.9	81.0	79.9	78.9	78.5	78.6	78.8	78.7	77.7	
		MCDB	83.8	84.9	87.3	88.3	87.6	84.9	83.5	83.2	83.7	84.6	84.8	84.2	
Mean Daily Temperature Range	5% DB	MDBR	12.6	11.7	10.7	11.2	10.3	8.7	8.5	8.1	8.9	10.0	10.9	12.2	
		MCDBR	13.2	12.5	11.3	11.6	11.1	10.5	10.2	9.4	10.1	11.0	11.9	12.8	
		MCWBR	5.2	4.7	4.3	5.0	4.5	4.4	4.6	3.8	4.0	4.2	4.7	5.0	
	5% WB	MCDBR	11.7	11.1	10.7	11.1	10.7	9.7	9.6	9.0	9.4	10.4	11.2	11.5	
		MCWBR	5.4	4.6	4.6	5.5	5.1	4.9	5.2	4.3	4.4	4.5	5.0	5.1	
Clear-Sky Solar Irradiance	taub	0.502	0.536	0.596	0.601	0.571	0.533	0.554	0.523	0.483	0.537	0.543	0.502		
	taud	2.108	2.020	1.893	1.902	1.991	2.105	2.030	2.132	2.263	2.079	2.040	2.139		
	Ebn at Noon	256	252	239	235	238	244	240	251	264	249	243	253		
	Edh at Noon	50	57	65	64	57	50	55	50	44	53	54	48		
All-Sky Solar Radiation	RadAvg	1778	1964	2011	1901	1666	1352	1358	1487	1652	1563	1498	1590		
	RadStd	63	65	82	71	120	122	119	128	146	91	84	77		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.72	N/A	N/A	+1.26	+0.86	+0.77	N/A	N/A	N/A	N/A

Nomenclature: See separate page